# Akshai Narayan Srinivasan

U.S Citizen | (425) 837-2049 | akshai.n.srinivasan@gmail.com | linkedin.com/in/akshaisrin | akshaisrin.live | github.com/akshaisrin

## Education

**University of Washington** **Seattle, WA**
*Dual Degree: BS in Computer Engineering, BS in Applied Mathematics* *June 2027*

**Relevant Coursework:** Machine Learning, Systems Programming, Digital Design, Data Structures & Parallelism, Hardware Software Interface, Discrete Mathematics & Probability, Dynamical Systems, Applied Linear Algebra, Differential Equations

## Technical Skills

**Languages**: Python, Java, C++, C, JavaScript, TypeScript, Dart
**Cloud & Databases**: AWS (Lambda, API Gateway, S3, DynamoDB, Bedrock), PostgreSQL, MongoDB, MySQL
**Frameworks & Tools**: ROS2, PyTorch, Isaac Lab, FastAPI, Claude API, Flask, WebSockets, Next.js, Stanford CoreNLP, Git

## Experience

**General Motors** **Sunnyvale, CA**
*Incoming Machine Learning Engineering Intern* *Summer 2026*

- Will be training state-of-the-art perception models for **Cruise's autonomous vehicles** on the AV Analysis team

**Washington Embodied Intelligence and Robotics Development Lab** **Seattle, WA**
*Undergraduate Research Assistant* *Feb 2026- Present*

- Developing sim-to-real **reinforcement learning policies** for robotic assembly tasks under **Dr. Abhishek Gupta** using **Isaac Lab** and **RSL-RL**, with WandB for experiment tracking
- Tuning **PPO** hyperparameters and reward weights to maximize dextrous manipulation success rates under domain randomization of object poses and dynamics parameters
- Implementing **linear curriculum learning, ADR, and LSTM**-based policy architectures to improve sim-to-real transfer and overall policy robustness

**Neptune Medical Inc.** **Burlingame, CA**
*Robotic Software Engineering Intern* *Jun 2025 – Sep 2025*

- Developed **control systems software** for **robotic endoscope** using **ROS** and **Python**, orchestrating communication between controllers, sensors, and actuators
- Built **C++ fault detection framework** for **haptic input device** with automatic failsafe protocols, preventing **unsafe robot operations** during live procedures
- Designed **tendon saturation algorithm** with integrated haptic feedback that dynamically generates resistive forces when tendons reach operational limits, preventing mechanical damage to endoscope

**DubHacks** **Seattle, WA**
*Director of Technology* *Jan 2025 – Present*

- Overseeing all software development and infrastructure for the **largest collegiate hackathon** in the Pacific Northwest (**1,200+** participants) using **FastAPI, Next.js, TypeScript, and React**
- Engineered **high-performance** judging platform with **OAuth2/JWT** authorization, leveraging **WebSockets** and event-driven architecture to achieve **sub-100ms latency** for **150+** concurrent sessions
- Architected scalable **MongoDB cluster** handling **376K edge requests** and **122K serverless function invocations** during peak hackathon traffic with **100% uptime** and zero data loss
- Built automated sentiment analysis engine to evaluate **1,800+** applications, reducing manual review time by **99%**

## Sample Projects (Portfolio: akshaisrin.live)

**BarelyAtWork** | *DeepGram, ElevenLabs, Third-Party APIs (Notion, Slack, G-Suite, Uber), Git* **Apr 2026**

- Developed a hands-free **agentic workflow builder** for the **Meta Raybans**, enabling users to integrate 3rd party tools to create custom voice-activated workflows from anywhere in the world
- Engineered an **agentic pipeline** that converts natural language into decomposed, step-by-step task representations before passing to an LLM, which resolves and executes each step using pre-integrated app context
- Implemented a **low-latency audio loop** using **Deepgram** for transcription, **Gemma** for intent parsing, and **ElevenLabs** for streaming spoken confirmation back through the glasses
- Integrated **6+ third-party services** via OAuth, enabling the agent to autonomously orchestrate **multi-service actions** (messaging, calendar updates, ride booking) from a single voice command